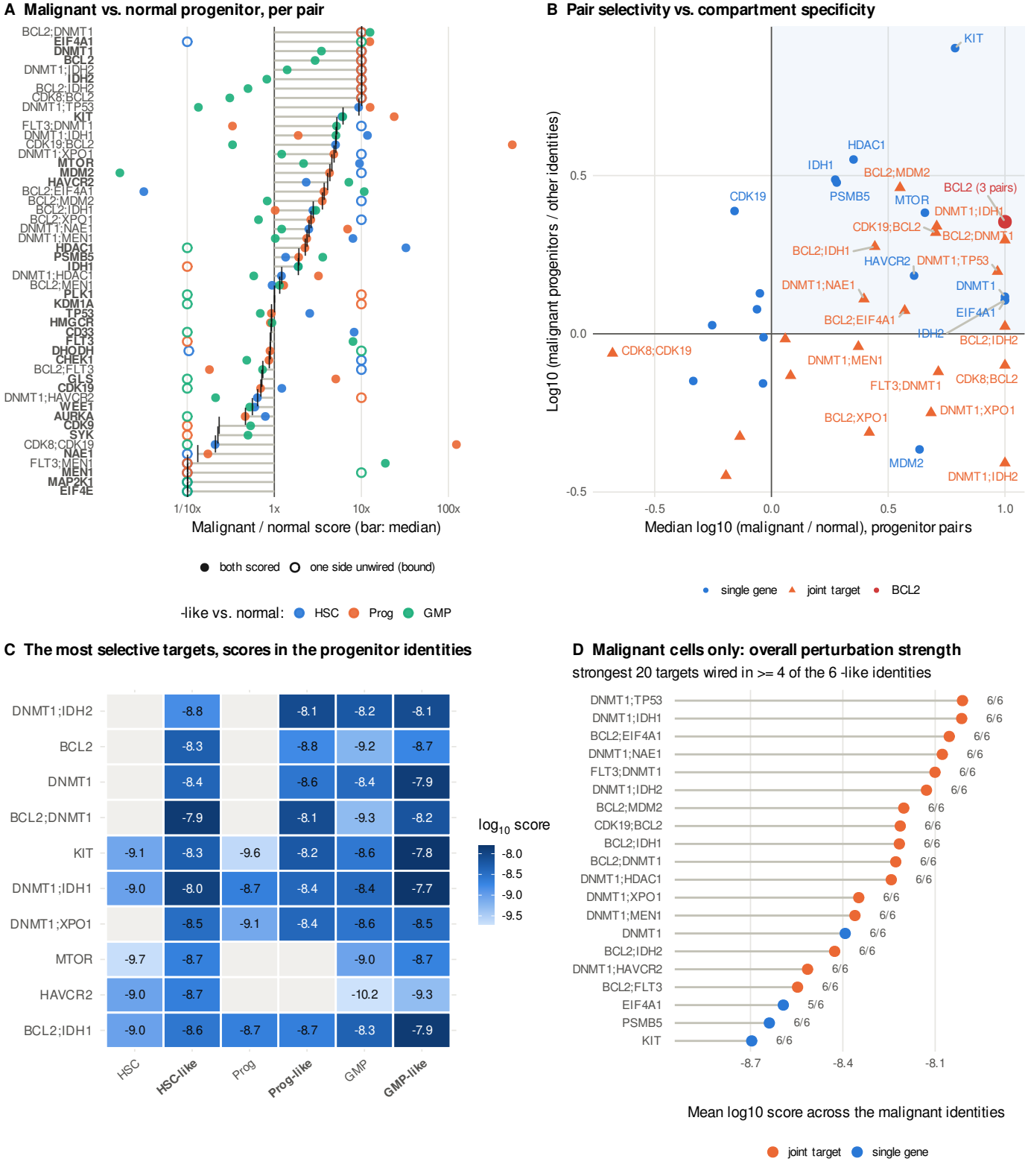
A Malignant vs. normal progenitor, per pair
BCL2;DNMT1
EIF4A1
DNMT1
BCL2
DNMT1;IDH2
IDH2
BCL2;IDH2
CDK8;BCL2
DNMT1;TP53
KIT
FLT3;DNMT1
DNMT1;IDH1
CDK19;BCL2
DNMT1;XPO1
MTOR
MDM2
HAVCR2
BCL2;EIF4A1
BCL2;MDM2
BCL2;IDH1
BCL2;XPO1
DNMT1;NAE1
DNMT1;MEN1
HDAC1
PSMB5
IDH1
DNMT1;HDAC1
BCL2;MEN1
PLK1
KDM1A
TP53
HMGCR
CD33
FLT3
DHODH
CHEK1
BCL2;FLT3
GLS
CDK19
DNMT1;HAVCR2
WEE1
AURKA
CDK9
SYK
CDK8;CDK19
NAE1
FLT3;MEN1
MEN1
MAP2K1
EIF4E
1/10x
1x
10x
100x
Malignant / normal score (bar: median)
both scored
one side unwired (bound)
-like vs. normal:
HSC
Prog
GMP
B Pair selectivity vs. compartment specificity
Log10 (malignant progenitors / other identities)
0.5
0.0
-0.5
-0.5
0.0
0.5
1.0
Median log10 (malignant / normal), progenitor pairs
KIT
HDAC1
IDH1
BCL2;MDM2
CDK19
PSMB5
MTOR
BCL2 (3 pairs)
DNMT1;IDH1
CDK19;BCL2
BCL2;DNMT1
BCL2;IDH1
HAVCR2
DNMT1;TP53
DNMT1;NAE1
DNMT1
BCL2;EIF4A1
EIF4A1
IDH2
BCL2;IDH2
CDK8;CDK19
DNMT1;MEN1
CDK8;BCL2
FLT3;DNMT1
DNMT1;XPO1
BCL2;XPO1
MDM2
DNMT1;IDH2
single gene
joint target
BCL2
C The most selective targets, scores in the progenitor identities
DNMT1;IDH2 -8.8 -8.1 -8.2 -8.1
BCL2 -8.3 -8.8 -9.2 -8.7
DNMT1 -8.4 -8.6 -8.4 -7.9
BCL2;DNMT1 -7.9 -8.1 -9.3 -8.2
KIT -9.1 -8.3 -9.6 -8.2 -8.6 -7.8
DNMT1;IDH1 -9.0 -8.0 -8.7 -8.4 -8.4 -7.7
DNMT1;XPO1 -8.5 -9.1 -8.4 -8.6 -8.5
MTOR -9.7 -8.7 -9.0 -8.7
HAVCR2 -9.0 -8.7 -10.2 -9.3
BCL2;IDH1 -9.0 -8.6 -8.7 -8.7 -8.3 -7.9
HSC
HSC-like
Prog
Prog-like
GMP
GMP-like
log10 score
-8.0
-8.5
-9.0
-9.5
D Malignant cells only: overall perturbation strength
strongest 20 targets wired in >= 4 of the 6 -like identities
DNMT1;TP53 6/6
DNMT1;IDH1 6/6
BCL2;EIF4A1 6/6
DNMT1;NAE1 6/6
FLT3;DNMT1 6/6
DNMT1;IDH2 6/6
BCL2;MDM2 6/6
CDK19;BCL2 6/6
BCL2;IDH1 6/6
BCL2;DNMT1 6/6
DNMT1;HDAC1 6/6
DNMT1;XPO1 6/6
DNMT1;MEN1 6/6
DNMT1 6/6
BCL2;IDH2 6/6
DNMT1;HAVCR2 6/6
BCL2;FLT3 6/6
EIF4A1 5/6
PSMB5 6/6
KIT 6/6
-8.7
-8.4
-8.1
Mean log10 score across the malignant identities
joint target
single gene